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Improved methane production from corn straw by microaerobic pretreatment with a pure bacteria system

Wanying Xu[a,1], Shanfei Fu[b,1], Zhiman Yang[a], Jun Lu[a], Rongbo Guo[a,*]

[a] Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, Qingdao, Shandong Province 266101, PR China
[b] School of Environment and Civil Engineering, Jiangnan University, Wuxi, Jiangsu Province 214122, PR China

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ABSTRACT

Thermophilic microaerobic pretreatment has been proved to be efficient in improving methane production of corn straw in previous studies. In this study, the effect of mesophilic (37 °C) microaerobic pretreatment using *Bacillus Subtilis* on the anaerobic digestion of corn straw was explored. Microaerobic pretreatment with a pure bacteria system was beneficial for the anaerobic digestion of corn straw, which obviously improved the methane yield. The maximum methane yield of 270.8 mL/g VS was obtained at the oxygen load of 5 mL/g VS, which was 17.35% higher than that of untreated group. Groups with mesophilic microaerobic pretreatment obtained higher glucose and VFAs concentrations, as well as higher peroxidase activities after 24 h pretreatment. In addition, the X-ray diffraction (XRD) analysis displayed the crystallinity indexes of pretreated groups were also decreased. Therefore, microaerobic pretreatment with a pure bacteria system (*Bacillus Subtilis*) is an efficient pretreatment method to enhance the anaerobic digestion efficiency of cellulosic biomass.

1. Introduction

Issues of energy crisis, environmental pollution and global warming promote the development of renewable energy (He et al., 2016b). Anaerobic digestion (AD) technology could not only meet the increasing of energy demands but also alleviate the problem of environmental pollution (Pöschl et al., 2010). Biogas production by AD provides one of the most cost-effective and efficient ways for producing renewable energy (Rouches et al., 2016). Lignocellulosic biomass is abundantly available bioresource for AD (Sindhu et al., 2016). However, limited by the composition and structure (Hendriks & Zeeman, 2009), the anaerobic digestion efficiency of lignocellulosic biomass was low (Wei et al., 2015). Therefore, during the anaerobic digestion of lignocellulosic biomass, a pretreatment process is usually needed to break down the structure barrier of lignocellulosic biomass and therefore improve methane production.

In general, methods of pretreatment are divided into several major categories, such as mechanical, chemical, thermal, thermo-chemical and biological pretreatments (Monlau et al., 2013). Compared with other pretreatment methods, biological pretreatments have the advantages of lower energy input, few AD inhibitors, environmental friendliness and less cost (Rabemanolontsoa & Saka, 2016). Lignin of lignocellulosic biomass blocked the accessibility of cellulose to hydrolytic enzymes (Kumar & Wyman, 2009). Hence, the key point of biological pretreatments is to break this structure by various enzymes to increase the hydrolysis rate of biomass.

The major hydrolysis enzymes of lignocellulosic biomass contain ligninase, cellulase, hemicellulase and pectinase (Van Dyk & Pletschke, 2012). Bacteria can produce various enzymes to accelerate the hydrolysis of lignocellulosic biomass. However, only some aerobic bacteria [e.g. *Rhodococcusjostii RHA1* (Ahmad et al., 2011), *Sphingobium* sp. *SYK-6* (Masai et al., 2007), *Bacillus ligniniphilus L1* and *Bacillus* sp. *BP-7* (Prim et al., 2003; Zhu et al., 2014)] can produce ligninolytic enzymes. Besides, bacteria have the advantages of small size, rapid propagation and high adaptability compared to fungus. Therefore, bacteria can be applied in the pretreatment process of digestion for hydrolyzing lignin and hemicellulose.

Micro-aeration in the digestion can improve the methane production (Botheju & Bakke, 2011; Jagadabhi et al., 2010). However, the oxygen is an important factor for digestion. In previous study, biogas slurry was served as inoculum with own microbial community in the microaerobic pretreatment, including anaerobic and facultative bacteria. The methane production during the AD of corn straw was improved for 16.24% after thermophilic (55 °C) microaerobic

* Corresponding author at: Shandong Industrial Engineering Laboratory of Biogas Production & Utilization, Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, Qingdao, Shandong Province 266101, PR China.
E-mail address: guorb@qibebt.ac.cn (R. Guo).
[1] These authors contributed same to this work.

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pretreatment. During thermophilic microaerobic pretreatment, the microbial community structure was changed to have a high relative abundance of class *Clostridia* (Fu et al., 2015c). The aerobic bacteria of *Bacillus licheniformis* was also used during the microaerobic pretreatment, which could increase methane production of algae AD about 17.6% (He et al., 2017). It decreased not only pretreatment time (60 h → 24 h) but also pretreatment temperature (55 °C → 37 °C). *Bacillus Subtilis* have many advantages, such as rapid propagation, high security and strong adaptability (Yang et al., 2017). In addition, *Bacillus subtilis* can produce dye-decolorizing peroxidases (DyPs) with high redox potential aromatic compounds (Santos et al., 2014). Therefore, Bacillus Subtilis is a potential inoculum during the microaerobic pretreatment of cellulosic substrate for anaerobic digestion.

In order to reduce energy consumption and cost, provide reference for large-scale application, pure bacteria (Bacillus Subtilis) was used as inoculum for microaerobic pretreatment in present study. The effect of microaerobic pretreatment with Bacillus Subtilis on the biogas production during the anaerobic digestion of corn straw was investigated and compared with biogas slurry. In addition, the crystallization degree of substrate and enzyme releasing during the pretreatment process was also studied. Furthermore, the hydrolysis of cellulosic structures was also studied.

2. Methods

2.1. Substrates, inoculum and bacteria

Air-dried corn straw, 7–8% moisture contents, was collected from cropland in Pingdu (Qingdao, Shandong province, China). After collection, samples were cut into particles (0.2–0.5 cm) by a hammer mill (FE130, Staida Co., Tianjing, China) and kept at ambient temperature (20–25 °C). Total solid (TS) content and volatile solid (VS) content of substrate were 91.5 ± 0.2% and 79.4 ± 2% (based on TS), respectively. The cellulose, hemicellulose and lignin contents of untreated corn straw were 41.3 ± 0.002%, 21.3 ± 0.003%, and 14.7 ± 0.003% (based on TS).

Biogas liquid was obtained from the Pingdu biogas plant (Qingdao, Shandong province, China). Total solid (TS) content and volatile solid (VS) content of the collected biogas liquid were 2.1 ± 0% and 71.2 ± 0.2% (based on TS), respectively. During anaerobic digestion, the inoculum to substrate ratio (VS/VS) was 1:2. Nutrient solution (NS) per liter contained 1 g NH_4Cl, 0.1 g $MgCl_2{\cdot}6H_2O$, 1 g NH_4Cl, 0.1 g NaCl, 0.05 g $CaCl_2{\cdot}2H_2O$, 2 g $K_2HPO_4{\cdot}3H_2O$, 2.6 g $NaHCO_3$, 0.5 g L- Cysteine hydrochloride anhydrous(Angelidaki et al., 2009) was used to adjust the ratio of inoculum to substrate.

Bacillus Subtilis was acquired from Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, China. Luria-Bertani (LB) solution medium contained 10 g tryptone, 5 g yeast extract, 10 g NaCl, and 1 L $H_2O$was used for *Bacillus Subtilis* cultivation. The *Bacillus Subtilis* were cultivated in sterile medium at 37 °C, 220 rpm for 24 h according to the growth curve (Yang et al., 2017).

2.2. Bacteria micro-aerobic pretreatment

Pretreatments were tested in closed vessels. The interaction was explored between *Bacillus Subtilis* and biogas liquid microbial community in micro-aerobic pretreatment of corn straw. Biogas liquid and bacteria solution were added in pretreatment system as the inoculum at different proportions: 1:0, 2:1, 1:1, 1:2, and 0:1 (v/v, biogas liquid/bacteria solution, marked as B1, B2, B3, B4, B5, respectively). The total volume of the pretreatment system was 100 mL. The inoculum volume was 68 mL according to the VS ratio of 2:1.The rest of volume was supplied by 32 mL of NS. All pretreatments were conducted in triplicates. Then, corn straw was also added into the bottles on the basis of VS ratio. The value of pH was adjusted to 7.4 by 2.0 M HCl or NaOH. Bottles were sealed with butyl rubber stoppers and vacuumized with vacuum pump. These bottles were also flushed with pure nitrogen to keep the air-pressure balance. Micro-aerobic pretreatment was followed by the oxygen loads of 5 mL/g $VS_{substrate}$ in line accordance with previous research (Fu et al., 2015b; He et al., 2017). Finally, bottles of batch pretreatment were put in a constant temperature air shaker at 37°C for 120 rpm about 24 h. During the micro-aerobic pretreatment process, the supernatants were sampled after 24 h for determination of volatile fatty acid (VFAs), peroxidase, carboxymethyl cellulase (CMCase) and glucose. Corn straw in every bottle was collected to measure crystallinity degree after pretreatment. A comparative group with the proportion of 1:0 was prepared again for control group before digestion. The control group, marked as C, was digested directly without bacterial and micro-aerobic pretreatment with the same of substrate, inoculum and NS.

2.3. Biochemical methane potential (BMP) assay

BMP experiments of pretreated and un-pretreated samples were also performed in closed vessels (Angelidaki et al., 2009). After bacteria micro-aerobic pretreatment, another 68 mL inoculum of biogas liquid was added to achieve $VS_{substrate}/VS_{inoculum}$ of 2. Then, NS was added to reach a total fermentation volume of 0.2 L. In addition, six background treatments were also prepared to eliminate the biogas and methane produced by bacterial culture. The background treatments included the same doses of inoculum, LB and NS corresponding to every treatment without substrates. The AD system was adjusted to pH 7.4 using 2.0 M NaOH or HCl. Afterwards, every treatment was sealed and vacuumized. All bottles were also flushed with pure argon to keep the anaerobic condition and air-pressure balance. All treatments were placed at 37 °C in constant temperature air shaker with 120 rpm (Li et al., 2009b). Biogas production and composition were measured by the experimental facility. All experiments were conducted in triplicates. Net CH_4 (mL/g VS) produced from corn straw was calculated by the following equation referring to algae AD researches (He et al., 2016a; Lü et al., 2013) (Eq. (1)).

$$CH_4\left(\frac{mL}{gVS}\right) = \frac{CH_4(mL,biomass\,and\,inoculum) - CH_4(mL,inoculum)}{biomass\,(gVS)} \quad (1)$$

2.4. Analytical methods

2.4.1. Basic analysis

TS and VS were analyzed according to the standard method (APHA, 2005). The volume of biogas was measured by water displacement method. Biogas composition was analyzed using a gas chromatograph (SP 6890, Shandong Lunan Inc., China) with a Porapak Q chromatographic column (180 cm × Ø3 mm) and a thermal conductivity detector (TCD). The argon was used as carrier gas in the process of biogas test.

Liquid samples were measured after centrifugation at 15,000*g* for 3 min and subsequent passing through 0.22 μm micro filter. The value of pH in the pretreatment process was test using pH meter (PB-21, Sartorius, Germany). Volatile Fatty Acid (VFAs) were analyzed by the gas chromatograph (GC-2014, Shimadzu, Japan) with WAX-DA column (30.0 m × Ø 0.32 mm × 0.50 μm) and flame ionization detector (FID) after filtrated samples were acidified by 2.0 M HCl to pH 3–4 (Shi et al., 2014). And the injection volume of liquid samples was 0.5μL. Glucose of liquid was tested by spectrophotometrically at 490 nm according to phenol-sulfuric acid spectrophotometric method (DuBois et al., 1956).

2.4.2. Enzyme analysis

Peroxidase activity of liquid in the pretreatment process was measured using spectrophotometrically at 440 nm according to the method of guaiacol (Li et al., 2009a). 1 unit (U) of peroxidase activity was

defined as the dosage of enzyme required to oxidize 1 nmol of guaiacol to tetra-o-methoxylenol in 1.0 min at 37 °C. Cellulase (CMCase) was determined by sodium carboxymethyl cellulose method at 540 nm (Ridge & Osborne, 1969). 1 U of CMCase was defined as the dosage of enzyme required to oxidize 4 mg/mL sodium carboxymethyl cellulose to release 1 μmol reducing sugar in 1.0 min.

2.4.3. Structural characterization analysis

The cellulose, hemicellulose and lignin contents of pretreated and fermented process were tested by Van Soest method (Van Soest, 1963) to observe morphological change of corn straw during pretreatment. Crystallinity index (CrI) of untreated and pretreated corn straw was measured by the wide angle X-ray diffraction (D8 Advance, Bruker, Germany). The operating conditions were 40 kV, 40 mA, and an angular range of from −100 to 168.The samples of straw were scanned in 2θ range from 5° to 60° at a scanning speed of 1.0°/min. Radiation (Cu Kα) is 1.5406 Å. The CrI was calculated by Eq. (2) (Hsu et al., 2010).

$$\text{Crytallinity Index(CrI)} = \frac{I_{crystallinity} - I_{amorphous}}{I_{crystallinity}} \times 100\% \quad (2)$$

where $I_{crystallinity}$ is the intensity for the crystalline portion of biomass at about 2 h = 22, $I_{amorphous}$ is the peak for the amorphous portion at about 2 h = 16.2.

2.5. Data analysis

Methane production characteristic was calculated according to modified Gompertz equation (Krishania et al., 2013) (Eq. (3)).

$$P(t) = P_m \times \exp\left\{-exp\left[\frac{R_m \times e}{P_m} \times (\lambda - t) + 1\right]\right\} \quad (3)$$

where P(t) is the accumulative methane production at time t (mL CH_4/gVS), P_m is the potential methane production (mL CH_4/g VS), R_m is the maximum methane production rate (mL/(g VS·d)), e is the base of the natural logarithms (2.7183), λ is the lag-phase time (d), t is the elapsed time (d). The values of λ, R_m and P(t) were obtained by nonlinear fitting using SPSS software.

The removals of lignin cellulose and hemicellulose during the pretreatment and digestion process were calculated as follows (Mustafa et al., 2017) (Eqs. (4)(6)).

$$\text{Lignin removal} = \frac{L_{initial} - L_{final}}{L_{initial}} \times 100\% \quad (4)$$

$$\text{Cellulose removal} = \frac{C_{initial} - C_{final}}{C_{initial}} \times 100\% \quad (5)$$

$$\text{Hemicellulose removal} = \frac{H_{initial} - H_{final}}{H_{initial}} \times 100\% \quad (6)$$

where L_{final}, C_{final}, H_{final} respectively are the fraction (g) of lignin cellulose and hemicellulose after pretreated corn straw while $L_{initial}$, $C_{initial}$, $H_{initial}$ are the corresponding fractions (g) of untreated corn straw.

Parameters were calculated using software of excel 2013. All figures in this study were analyzed and plotted by Origin 9.0 software.

3. Results and discussion

3.1. VFA concentrations and compositions

The concentrations and composition of VFAs among pretreated and un-pretreated groups are demonstrated in Fig. 1. After 24 h of pretreatment, the concentrations of VFAs showed obvious improvement compared to un-pretreated group C. The VFAs concentration of B1, B2, B3, B4 and B5 was 1414.45, 2491.04, 3868.94, 2824.56 and 4185.72 mg/L, respectively. The high VFAs concentrations would be

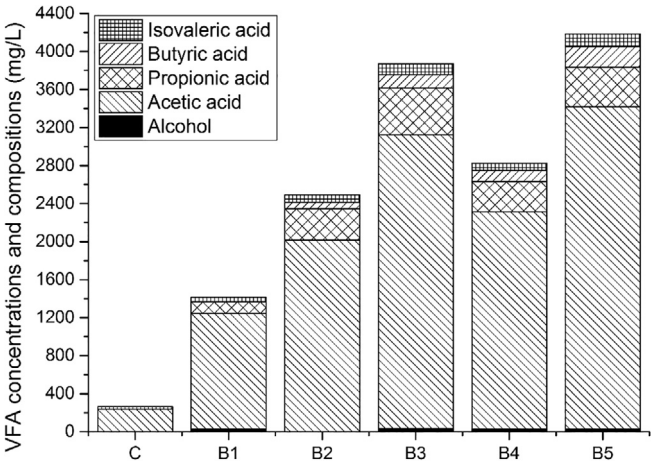


Fig. 1. Concentrations and composition of VFAs after 24 h pretreatment.

beneficial for subsequent methanogenesis. The highest VFAs concentration was obtained from B5 group with the proportions of 0:1 (v/v, biogas liquid/bacteria solution), which was 16 times higher than that of untreated group. The concentrations of alcohol, acetic acid, propionic acid, butyric acid and valeric acid of B5 were 30.13, 3390.91, 413.82, 215.36 and 135.50 mg/L, respectively. Acetate and butyrate are the key components of VFAs in the hydrolysis process to improve methane production (Wang et al., 1999). Thus, the accumulated methane production of B5 group was higher than other groups with more the concentrations of acetate and butyrate (showed in Fig. 1). In contrast, propionate and valerate are inhibited factors for methane generation (Xu et al., 2014). B3 produced more propionate, which lead to lower methane improvement rate of B3 group compared to other pretreated groups. The VFAs of B5 were 3 times higher than B1, but methane improvement rate was only 3% more than B1.

3.2. Glucose concentration

Glucoses usually produced from the hydrolysis of cellulose and hemicelluloses structure of cellulosic biomass (Dos Santos Castro et al., 2014). Then the glucose was changed into VFAs in the acidification process (Oyiwona et al., 2017). Therefore, the change of glucose concentrations reflected biomass transformation in the pretreatment process. The variation of glucose concentration was demonstrated in Fig. 2. The glucose concentration with *Bacillus Subtilis* liquid was higher at time 0. The glucose concentration increased with the increase in the solution proportion of *Bacillus Subtilis*. But only glucose concentration

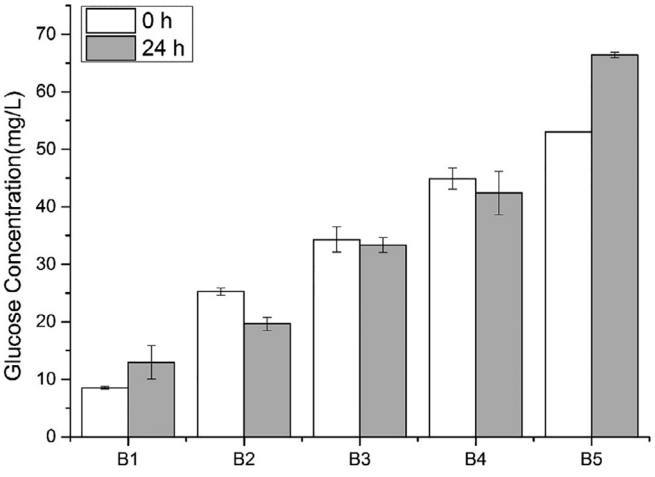


Fig. 2. Concentrations and of glucose after 24 h pretreatment.

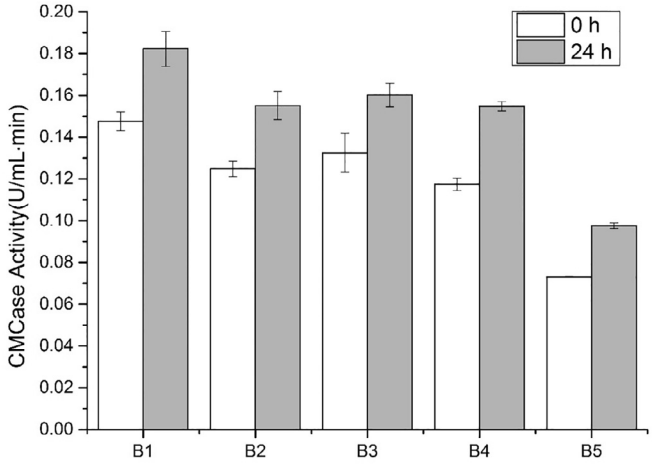


Fig. 3. CMCase activity of pretreated groups.

of B1 and B5 were promoted after 24 h pretreatment, and enhancement rate was 52% and 25% respectively. It could be concluded that glucose was production in the pretreatment process from straw by biogas slurry microbial flora and *Bacillus Subtilis*.

3.3. Enzyme activity

3.3.1. Cellulase activity

CMCase activity reflects activity of endow β-1.4-glucan for hydrolyzing cellulose (Macris et al., 1987). The CMCase of pretreated groups is shown in Fig. 3. The CMCase of every group is improved after 24 h pretreatment. The CMCase enhancement rate of B1, B2, B3, B4 and B5 is 23.45%, 24.33%, 20.91%, 31.82% and 33.57%, respectively. The maximum of CMCase enhancement rate is B5 with pure bacterium inoculum. But B1, B2, B3 and B4 have higher CMCase activity at time 0. Finally, the CMCase activity of B1, B2, B3, B4 and B5 is 0.18, 0.16, 0.16, 0.15 and 0.10 U/mL min after 24 h bacteria microaerobic pretreatment. Therefore, cellulose hydrolysis ability of B1 is higher than other groups. And biogas slurry microflora was tending to hydrolyze cellulose.

3.3.2. Lignin enzyme activity

Peroxidase is an oxidation-reductase secreted by microorganism or plant. It can hydrolyze aromatic compounds, like lignin (Monlau et al., 2013; Pollegioni et al., 2015). Peroxidase activity of pretreated is demonstrated in Fig. 4. There are no peroxidase activities at time 0 in B1, B2, B3, B4 and B5. However, B4 and B5 produce peroxidase activity at time 24 after bacteria microaerobic pretreatment. The peroxidase activity of B4 and B5 are 0.47 and 4.24 U/mL min, respectively. It draws a conclusion that during the pretreatment process, lignin structure could be partly destroyed by *Bacillus Subtilis*. This result is lower than the previous study with 15 U/mg protein peroxidase activity because of different units, pH value and substrate (Min et al., 2015). However, *Bacillus Subtilis* have higher peroxidase activity in condition of lower pH (3–5) and higher temperature (50–55 °C) (Min et al., 2015; Santos et al., 2014) reached up to 66.8 U/mg proteins (Min et al., 2015). Therefore, pretreated time and temperature can be used as important factors for bacteria microaerobic pretreatment of cellulosic biomass in the future.

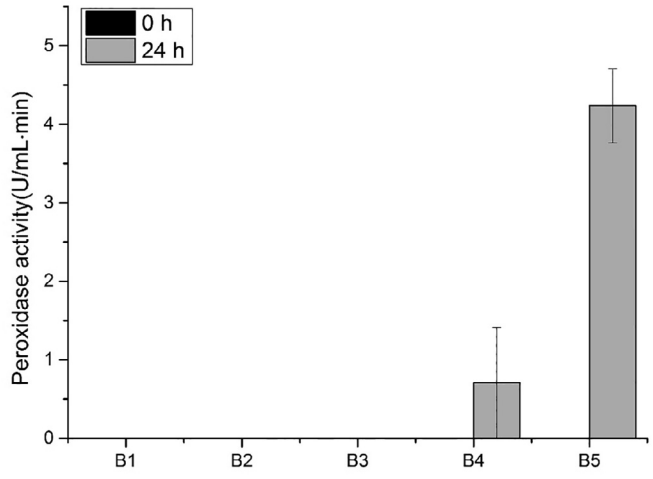


Fig. 4. Peroxidase activity of pretreated groups.

3.4. Crystallinity degree analysis

It is broadly reported that amorphous cellulose of biomass is easy attacked by various enzymes compared to highly crystalline cellulose (Teeri, 1997). Microorganism can attack highly crystalline cellulose to decrease crystallinity degree of biomass (Gupta et al., 2016; Nakashima et al., 2014). Therefore, crystallinity degree is an important factor to reflect enzyme hydrolysis during the pretreatment process. In this study, the crystallinity degree of corn straw was analyzed by XRD. The results are shown in Table 1. After microaerobic pretreatment, the crystallinities of pretreated corn straw were lower than that of untreated group. The lower crystallinity index revealed more crystalline cellulose was destroyed (Kuo and Lee, 2009). The minimum crystallinity of about 25.58 was obtained from B5. Hence, it could be concluded that *Bacillus Subtilis*is beneficial for the microaerobic pretreatment of corn straw. Thus, aerobic bacteria are regarded as inoculum for microaerobic pretreatment that could improve the methane production of corn straw anaerobic digestion.

3.5. Degradation of lignin and hemicellulose

All of lignin, cellulose and hemicellulose are hydrolyzed to some extent after microaerobic pretreatment. The hydrolysis rate is shown in Fig. 5. Except B3, B1, B2 and B4 with biogas slurry have higher cellulase hydrolysis rate about 5.0–7.5%. It reflected biogas slurry mainly contains bacteria with the ability to hydrolyze cellulose, which was in keeping with the CMCase activity result. Lignin of all groups was hydrolyzed, and the scope of hydrolysis rate was 17–23%. The max hydrolysis rate was obtained from B5, which was also kept in line with peroxidase activity result. Peroxidase activities in B1, B2 and B3 were lower than testing level. In addition, B4 and B5 showed stronger ability of hemicellulose hydrolysis. The hydrolysis rate of hemicellulose was 17.7% and 18.1%, respectively.

3.6. Methane production

3.6.1. Accumulative methane production in AD of corn straw

The accumulative methane productions of pretreated and un-pretreated groups after 51 days' anaerobic digestion were shown in Fig. 6. The methane yields increased sharply from 0 to 17 days. The final methane productions of C, B1, B2, B3, B4, and B5 groups were 230.7 ± 5.0, 263.9 ± 4.0, 246.4 ± 0.1, 241.5 ± 10.0, 258.9 ± 2.0 and 270.8 ± 7.0 mL/gVS, respectively. The methane yields of all

Table 1
Crystallinity indexes of different percentage inoculum.

Groups	Crystallinity index	Relative change (%, relative to C)
C	29.70	0.00
B1	27.17	−2.53
B2	29.13	−0.58
B3	27.69	−2.01
B4	25.86	−3.84
B5	25.58	−4.12

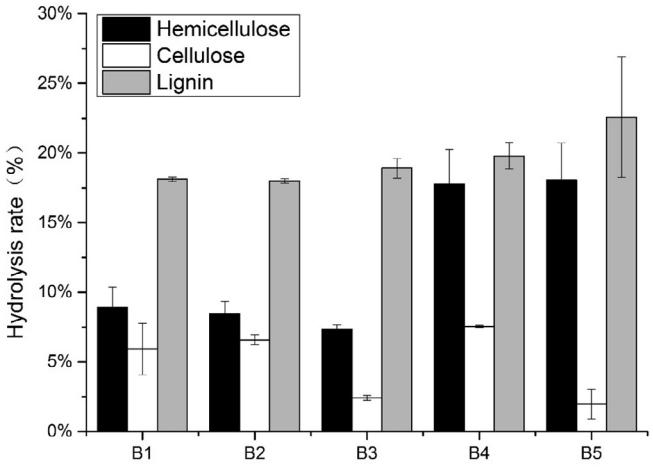


Fig. 5. Biomass hydrolysis rate after 24 h pretreatment.

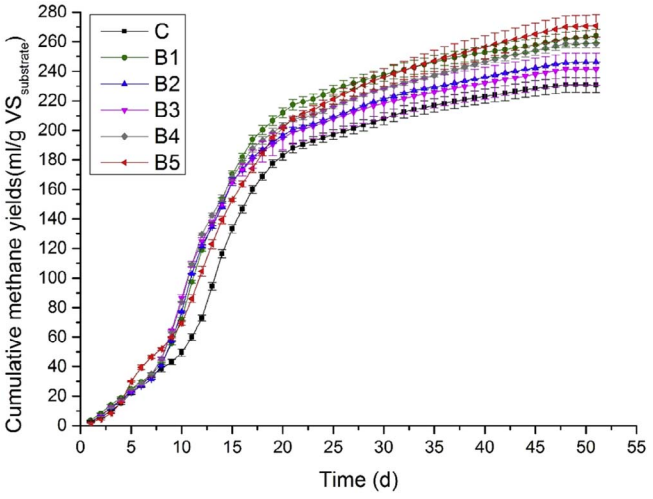


Fig. 6. The cumulative methane productions of pretreated and non-pretreated corn straw biomass.

pretreated groups were higher than control group (C). The maximum cumulative methane yield was obtained from B5, which was 17.35% higher than that of group C. The accumulated methane production of group B1 was 14.41% higher than that of group C, but lower than the recent study of 16.24% at same oxygen load and pretreated inoculum (Fu et al., 2015a). These might be caused by different pretreated temperature (37 °C or 55 °C). The thermophilic bacterial population in the pretreated phase could accelerate the hydrolysis process at 55 °C (Dumas et al., 2010). Thus, aerobic bacteria instead of biogas liquid microflora as pretreated inoculum can decrease pretreated temperature and therefore reducing energy consumption. The methane increasing rate of B5 group was similar to *B. Licheniformis* of 17.6% at the same oxygen load, pretreated temperature and pretreated time (He et al., 2017). But, the methane yield of Chlorella sp was enhanced by 22.7% after 60 h anaerobic pretreatment of *B. licheniformis* without micro-aerobic pretreatment (He et al., 2016a). The pretreated time of lignocellulosic biomass should be study in degree at micro-aerobic condition. In addition, the methane improving rate was the worst when the ratio of biogas liquid and bacteria solution was 1:1.

3.6.2. Gompertz model analysis

The accumulated methane productions of all groups were fitted with the modified Gompertz model, results were shown in Table 2. R square parameters of the fitted data were above 0.99. The maximum methane yield of simulated results was obtained from B5, which was also in line

Table 2
Parameters of Gompertz equation fitting experimental data.

Groups	P_m(mL CH_4/gVS)	R_m(mL/(g VS·d))	λ(d)	R^2
C	225.67	13.617	5.53	0.993
B1	253.72	16.426	4.89	0.994
B2	236.32	15.213	4.51	0.992
B3	232.15	15.034	4.18	0.993
B4	247.04	15.084	4.14	0.990
B5	262.84	13.245	3.90	0.995

with experimental data. Methane yield of all the pretreated groups were higher than that of control group. The methane production rates were also higher in pretreated groups compared with C in accordance with the previous studies (He et al., 2017; Liu et al., 2015). In AD phase, the lag-phase time (λ) of pretreated groups were lower than C group in agreement with the results reported by Fu et al. (2015a). And the higher bacteria solution proportion was, the shorter lag-phase time obtained. The result was in contrast to the previous study (He et al., 2016a). Because *Bacillus Subtilis* was a kind of bacteria with stronger environment adaption (Earl et al., 2008), and population size also affected system stability and environmental suitability of microorganism (Giovannoni and Stingl, 2005). Thus, it can be concluded that *Bacillus Subtilis* reduced the lag-phase time of AD with the increase of bacteria solution proportion.

4. Conclusion

Aerobic bacteria (*Bacillus Subtilis*) is an efficient inoculum instead of biogas slurry for enhancing microaerobic pretreatment performance during the anaerobic digestion of cellulosic biomass. Compared with biogas slurry, the pure bacteria system performed better in removing lignin and reducing crystallinity index. Therefore, microaerobic pretreatment with a pure bacteria system was more beneficial for the anaerobic digestion of corn straw, which obviously improved the methane yields. The higher bacteria solution proportion was, the shorter lag-phase time obtained. Microaerobic pretreatment with a pure bacteria system could decrease the lag-phase time and pretreated temperature.

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References

Ahmad, M., Roberts, J.N., Hardiman, E.M., Singh, R., Eltis, L.D., Bugg, T.D., 2011. Identification of DypB from Rhodococcusjostii RHA1 as a lignin peroxidase. Biochemistry 50 (23), 5096–5107.

Angelidaki, I., Alves, M., Bolzonella, D., Borzacconi, L., Campos, J., Guwy, A., Kalyuzhnyi, S., Jenicek, P., Van Lier, J., 2009. Defining the biomethane potential (BMP) of solid organic wastes and energy crops: a proposed protocol for batch assays. Water Sci. Technol. 59 (5), 927–934.

APHA, 2005. Standard methods for the examination of water and wastewater, APHA.

Botheju, D., Bakke, R., 2011. Oxygen effects in anaerobic digestion – a review. Model. Identif. Control 411 (1), 1–19.

Dos Santos Castro, L., Pedersoli, W.R., Antoniêto, A.C.C., Steindorff, A.S., Silva-Rocha, R., Martinez-Rossi, N.M., Rossi, A., Brown, N.A., Goldman, G.H., Faça, V.M., 2014. Comparative metabolism of cellulose, sophorose and glucose in Trichodermareesei using high-throughput genomic and proteomic analyses. Biotechnol. Biofuels 7 (1), 41.

DuBois, M., Gilles, K.A., Hamilton, J.K., Rebers, P.t., Smith, F., 1956. Colorimetric method

for determination of sugars and related substances. Anal. Chem. 28 (3), 350–356.
Dumas, C., Perez, S., Paul, E., Lefebvre, X., 2010. Combined thermophilic aerobic process and conventional anaerobic digestion: effect on sludge biodegradation and methane production. Bioresour. Technol. 101 (8), 2629–2636.
Earl, A.M., Losick, R., Kolter, R., 2008. Ecology and genomics of Bacillus subtilis. Trends Microbiol. 16 (6), 269–275.
Fu, S.-F., He, S., Shi, X.-S., Katukuri, N.R., Dai, M., Guo, R.-B., 2015a. The chemical properties and microbial community characterization of the thermophilic micro-aerobic pretreatment process. Bioresour. Technol. 198, 497–502.
Fu, S.-F., Shi, X.-S., Xu, X.-H., Wang, C.-S., Wang, L., Dai, M., Guo, R.-B., 2015b. Secondary thermophilic microaerobic treatment in the anaerobic digestion of corn straw. Bioresour. Technol. 186, 321–324.
Fu, S.F., Wang, F., Yuan, X.Z., Yang, Z.M., Luo, S.J., Wang, C.S., Guo, R.B., 2015c. The thermophilic (55°C) microaerobic pretreatment of corn straw for anaerobic digestion. Bioresour. Technol. 175C (7), 203–208.
Giovannoni, S.J., Stingl, U., 2005. Molecular diversity and ecology of microbial plankton. Nature 437 (7057), 343–348.
Gupta, V.K., Kubicek, C.P., Berrin, J.-G., Wilson, D.W., Couturier, M., Berlin, A., EdivaldoFilho, X., Ezeji, T., 2016. Fungal enzymes for bio-products from sustainable and waste biomass. Trends Biochem. Sci. 41 (7), 633–645.
He, S., Fan, X., Katukuri, N.R., Yuan, X., Wang, F., Guo, R.-B., 2016a. Enhanced methane production from microalgal biomass by anaerobic bio-pretreatment. Bioresour. Technol. 204, 145–151.
He, S., Fan, X., Luo, S., Katukuri, N.R., Guo, R., 2017. Enhanced the energy outcomes from microalgal biomass by the novel biopretreatment. Energy Convers. Manage. 135, 291–296.
He, Y., Xu, Y., Pang, Y., Tian, H., Wu, R., 2016b. A regulatory policy to promote renewable energy consumption in China: review and future evolutionary path. Renew. Energy 89, 695–705.
Hendriks, A., Zeeman, G., 2009. Pretreatments to enhance the digestibility of lignocellulosic biomass. Bioresour. Technol. 100 (1), 10–18.
Hsu, T.-C., Guo, G.-L., Chen, W.-H., Hwang, W.-S., 2010. Effect of dilute acid pretreatment of rice straw on structural properties and enzymatic hydrolysis. Bioresour. Technol. 101 (13), 4907–4913.
Jagadabhi, P.S., Kaparaju, P., Rintala, J., 2010. Effect of micro-aeration and leachate replacement on COD solubilization and VFA production during mono-digestion of grass-silage in one-stage leach-bed reactors. Bioresour. Technol. 101 (8), 2818.
Krishania, M., Vijay, V., Chandra, R., 2013. Methane fermentation and kinetics of wheat straw pretreated substrates co-digested with cattle manure in batch assay. Energy 57, 359–367.
Kumar, R., Wyman, C.E., 2009. Effect of xylanase supplementation of cellulase on digestion of corn stover solids prepared by leading pretreatment technologies. Bioresour. Technol. 100 (18), 4203.
Kuo, C.-H., Lee, C.-K., 2009. Enhancement of enzymatic saccharification of cellulose by cellulose dissolution pretreatments. Carbohydr. Polym. 77 (1), 41–46.
Li, X., Jia, R., Li, P., Ang, S., 2009a. Response surface analysis for enzymatic decolorization of Congo red by manganese peroxidase. J. Mol. Catal. B: Enzym. 56 (1), 1–6.
Li, X., Ma, H., Wang, Q., Matsumoto, S., Maeda, T., Ogawa, H.I., 2009b. Isolation, identification of sludge-lysing strain and its utilization in thermophilic aerobic digestion for waste activated sludge. Bioresour. Technol. 100 (9), 2475–2481.
Liu, X., Zicari, S.M., Liu, G., Li, Y., Zhang, R., 2015. Pretreatment of wheat straw with potassium hydroxide for increasing enzymatic and microbial degradability. Bioresour. Technol. 185, 150–157.
Lü, F., Ji, J., Shao, L., He, P., 2013. Bacterial bioaugmentation for improving methane and hydrogen production from microalgae. Biotechnol. Biofuels 6 (1), 92.
Macris, B., Kekos, D., Evangelidou, X., Galiotou-Panayotou, M., Sodis, P., 1987. Solid state fermentation of straw withNeurosporacrassa for CMCase and β-glucosidase production. Biotechnol. Lett. 9 (9), 661–664.
Masai, E., Katayama, Y., Fukuda, M., 2007. Genetic and biochemical investigations on bacterial catabolic pathways for lignin-derived aromatic compounds. Biosci. Biotechnol. Biochem. 71 (1), 1–15.
Min, K., Gong, G., Woo, H.M., Kim, Y., Um, Y., 2015. A dye-decolorizing peroxidase from Bacillus subtilis exhibiting substrate-dependent optimum temperature for dyes and β-ether lignin dimer. Sci. R. 5, 8245.
Monlau, F., Barakat, A., Trably, E., Dumas, C., Steyer, J.-P., Carrère, H., 2013. Lignocellulosic materials into biohydrogen and biomethane: impact of structural features and pretreatment. Crit. Rev. Environ. Sci. Technol. 43 (3), 260–322.
Mustafa, A.M., Poulsen, T.G., Xia, Y., Sheng, K., 2017. Combinations of fungal and milling pretreatments for enhancing rice straw biogas production during solid-state anaerobic digestion. Bioresour. Technol. 224, 174–182.
Nakashima, K., Endo, K., Shibasaki-Kitakawa, N., Yonemoto, T., 2014. A fusion enzyme consisting of bacterial expansin and endoglucanase for the degradation of highly crystalline cellulose. RSC Adv. 4 (83), 43815–43820.
Oyiwona, G.E., Ogbonna, J., Anyanwu, C.U., Ishizaki, S., Kimura, Z.-I., Okabe, S., 2017. Oxidation of glucose by syntrophic association between Geobacter and hydrogenotrophic methanogens in microbial fuel cell. Biotechnol. Lett. 39 (2), 253–259.
Pollegioni, L., Tonin, F., Rosini, E., 2015. Lignin-degrading enzymes. FEBS J. 282 (7), 1190–1213.
Pöschl, M., Ward, S., Owende, P., 2010. Evaluation of energy efficiency of various biogas production and utilization pathways. Appl. Energy 87 (11), 3305–3321.
Prim, N., Pastor, F., Diaz, P., 2003. Biochemical studies on cloned Bacillus sp. BP-7 phenolic acid decarboxylase PadA. Appl. Microbiol. Biotechnol. 63 (1), 51–56.
Rabemanolontsoa, H., Saka, S., 2016. Various pretreatments of lignocellulosics. Bioresour. Technol. 199, 83–91.
Ridge, I., Osborne, D.J., 1969. Cell growth and cellulases: regulation by ethylene and indole-3-acetic acid in shoots of Pisumsativum. Nature 223 (5203), 318–319.
Rouches, E., Herpoël-Gimbert, I., Steyer, J., Carrere, H., 2016. Improvement of anaerobic degradation by white-rot fungi pretreatment of lignocellulosic biomass: a review. Renew. Sustain. Energy Rev. 59, 179–198.
Santos, A., Mendes, S., Brissos, V., Martins, L.O., 2014. New dye-decolorizing peroxidases from Bacillus subtilis and Pseudomonas putida MET94: towards biotechnological applications. Appl. Microbiol. Biotechnol. 98 (5), 2053–2065.
Shi, J., Xu, F., Wang, Z., Stiverson, J.A., Yu, Z., Li, Y., 2014. Effects of microbial and non-microbial factors of liquid anaerobic digestion effluent as inoculum on solid-state anaerobic digestion of corn stover. Bioresour. Technol. 157, 188–196.
Sindhu, R., Binod, P., Pandey, A., 2016. Biological pretreatment of lignocellulosic biomass–An overview. Bioresour. Technol. 199, 76–82.
Teeri, T.T., 1997. Crystalline cellulose degradation: new insight into the function of cellobiohydrolases. Trends Biotechnol. 15 (5), 160–167.
Van Dyk, J., Pletschke, B., 2012. A review of lignocellulose bioconversion using enzymatic hydrolysis and synergistic cooperation between enzymes—factors affecting enzymes, conversion and synergy. Biotechnol. adv. 30 (6), 1458–1480.
Van Soest, P.J., 1963. Use of detergents in the analysis of fibrous feeds. 2. A rapid method for the determination of fiber and lignin. J. Assoc. Offic. Agric. Chem. 46, 829–835.
Wang, Q., Kuninobu, M., Ogawa, H.I., Kato, Y., 1999. Degradation of volatile fatty acids in highly efficient anaerobic digestion. Biomass Bioenergy 16 (6), 407–416.
Wei, Y., Li, X., Yu, L., Zou, D., Yuan, H., 2015. Mesophilic anaerobic co-digestion of cattle manure and corn stover with biological and chemical pretreatment. Bioresour. Technol. 198, 431–436.
Xu, S., Selvam, A., Wong, J.W., 2014. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. Waste Manage. 34 (2), 363–369.
Yang, S., Du, G., Chen, J., Kang, Z., 2017. Characterization and application of endogenous phase-dependent promoters in Bacillus subtilis. Appl. Microbiol. Biotechnol. 101 (10), 4151–4161.
Zhu, D., Tanabe, S.-H., Xie, C., Honda, D., Sun, J., Ai, L., 2014. Bacillus ligniniphilus sp. nov., an alkaliphilic and halotolerant bacterium isolated from sediments of the South China Sea. Int. J. Syst. Evol. Microbiol. 64 (5), 1712–1717.